

Supplemental Online Content

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eTable 1. Sensitivity Analyses of the Primary Outcome

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This supplementary material has been provided by the authors to give readers additional information about their work.

Supplementary Table 1: Sensitivity analyses of the primary outcome

	Sildenafil Citrate (N=1634)	Placebo (N=1641)	Sildenafil Citrate vs Placebo	
	no./total no. (%)	no./total no. (%)	Relative Risk (95% CI)	P-value
Composite primary endpoint with cord artery pH <7 imputed (10 components, GEE) ^{†§}				
Yes	85/1625 (5.2%)	87/1625 (5.4%)	1.01 (0.75 - 1.36)	0.943
No	1540/1625 (94.8%)	1538/1625 (94.6%)		
Missing	9	16		
Cord artery pH < 7.0 (after imputation, GEE) ^{†§}				
Yes	14/1629 (0.9%)	8/1637 (0.5%)	1.62 (0.68 - 3.85)	0.276
No	1211/1629 (74.3%)	1154/1637 (70.5%)		
Not Done	404/1629 (24.8%)	475/1637 (29.0%)		
Missing	5	4		
Composite primary endpoint (9 components, GEE) [†]				
Yes	75/1625 (4.6%)	80/1625 (4.9%)	0.96 (0.70 - 1.31)	0.796
No	1550/1625 (95.4%)	1545/1625 (95.1%)		
Missing	9	16		
Composite primary endpoint (10 components, GLM) ^{‡¶}				
Yes	83/1625 (5.1%)	84/1625 (5.2%)	0.99 (0.74 - 1.33)	0.937
No	1542/1625 (94.9%)	1541/1625 (94.8%)		
Missing	9	16		
Composite primary endpoint (10 components, GLM, adjusted for Cord artery Not Done) ^{‡¶¥}			0.98 (0.73 - 1.31)	0.882
Composite primary endpoint with cord artery pH <7 imputed (10 components, GLM, adjusted for Cord artery Not Done) ^{‡§¥}			0.97 (0.73 - 1.30)	0.844

*18 sets of twins

[†]GEE log-binomial model, adjusts for clustering due to twins

[‡]GLM log-binomial, does not adjust for clustering

[§]Additional cord artery pH <7 events imputed for Not Done category based on observed pooled event rate (where cord artery testing was done, and no other outcome component was observed). The imputation was stratified by twin status.

[¶]Not Done for Cord Artery pH<7 assumed to be No.

[¥]Model includes a Cord Artery testing 'Not Done' indicator variable to adjust for imputation of missing data

Supplementary Table 2: Primary, secondary and tertiary outcomes for Mater Mother’s Hospital vs. other sites

	Sildenafil		Placebo		Sildenafil vs Placebo*		
	N	n (%)	N	n (%)	Relative Risk (95% CI)	P	P interaction
Primary outcome							
Mater Mother’s Hospital	883	16 (1.8%)	887	24 (2.7%)	0.67 (0.36 - 1.25)	0.209	0.158
Other sites	742	67 (9.0%)	738	60 (8.1%)	1.11 (0.80 - 1.55)	0.537	
Secondary outcomes							
Intrapartum stillbirth							
Mater Mother’s Hospital	883	0	887	0	-		
Other Sites	746	0	750	0	-		
28-day neonatal death							
Mater Mother’s Hospital	883	0	887	0	-		
Other Sites	742	0	740	0	-		
Apgar score <4 at 5 minutes							0.676
Mater Mother’s Hospital	883	1 (0.1%)	887	1 (0.1%)	1.00 (0.06 - 16.03)	0.997	
Other Sites	746	4 (0.5%)	750	2 (0.3%)	2.01 (0.37 - 10.94)	0.419	
Cord artery pH < 7.0 [†]							0.315
Mater Mother’s Hospital	883	2 (0.2%)	887	2 (0.2%)	1.00 (0.14 - 7.12)	0.996	
Other Sites	746	10 (1.3%)	750	3 (0.4%)	3.35 (0.93 - 12.13)	0.065	
Neonatal encephalopathy							
Mater Mother’s Hospital	883	0 (0.0%)	887	0 (0.0%)			
Other Sites	742	1 (0.1%)	738	1 (0.1%)	0.99 (0.06 - 15.87)	0.997	
Neonatal seizures within 28 days							
Mater Mother’s Hospital	883	0 (0.0%)	887	0 (0.0%)			
Other Sites	742	2 (0.3%)	738	1 (0.1%)	1.99 (0.18 - 21.89)	0.574	
Neonatal unit admission >48 hours							0.931
Mater Mother’s Hospital	883	11 (1.2%)	887	13 (1.5%)	0.85 (0.38 - 1.89)	0.690	
Other Sites	746	39 (5.2%)	750	48 (6.4%)	0.82 (0.54 - 1.23)	0.334	

	Sildenafil		Placebo		Sildenafil vs Placebo*		
	N	n (%)	N	n (%)	Relative Risk (95% CI)	P	P interaction
Neonatal respiratory support >4hrs							0.015
Mater Mother’s Hospital	883	5 (0.6%)	887	17 (1.9%)	0.30 (0.11 - 0.80)	0.016	
Other Sites	746	32 (4.3%)	750	30 (4.0%)	1.07 (0.66 - 1.75)	0.779	
Persistent pulmonary hypertension of the newborn							
Mater Mother’s Hospital	883	0 (0.0%)	887	1 (0.1%)			
Other Sites	746	1 (0.1%)	750	2 (0.3%)	0.50 (0.05 - 5.53)	0.574	
Meconium aspiration							0.214
Mater Mother’s Hospital	883	1 (0.1%)	887	2 (0.2%)	0.50 (0.05 - 5.53)	0.574	
Other Sites	746	8 (1.1%)	750	3 (0.4%)	2.68 (0.71 - 10.07)	0.144	
Cesarean or IVB for fetal distress†							0.591
Mater Mother’s Hospital	883	169 (19.1%)	887	157 (17.7%)	1.08 (0.89 - 1.32)	0.435	
Other Sites	746	174 (23.3%)	750	150 (20.0%)	1.17 (0.96 - 1.42)	0.119	
Tertiary outcomes							
Cesarean for fetal distress (primary indication)							0.718
Mater Mother’s Hospital	883	73 (8.3%)	887	69 (7.8%)	1.06 (0.78 - 1.46)	0.705	
Other Sites	746	69 (9.2%)	750	60 (8.0%)	1.16 (0.83 - 1.61)	0.390	
IVB for fetal distress (primary indication)							0.726
Mater Mother’s Hospital	883	96 (10.9%)	887	88 (9.9%)	1.10 (0.83 - 1.44)	0.512	
Other Sites	746	105 (14.1%)	750	90 (12.0%)	1.17 (0.90 - 1.53)	0.234	
Cesarean for other indications							0.815
Mater Mother’s Hospital	883	148 (16.8%)	887	133 (15.0%)	1.12 (0.90 - 1.39)	0.310	
Other Sites	746	143 (19.2%)	750	124 (16.5%)	1.16 (0.93 - 1.44)	0.184	
IVB for other indications							0.559
Mater Mother’s Hospital	883	54 (6.1%)	887	61 (6.9%)	0.89 (0.62 - 1.27)	0.516	
Other Sites	746	40 (5.4%)	750	53 (7.1%)	0.76 (0.51 - 1.13)	0.174	
Emergency operative birth (fetal distress was a factor)							0.927

	Sildenafil		Placebo		Sildenafil vs Placebo*		
	N	n (%)	N	n (%)	Relative Risk (95% CI)	P	P interaction
Mater Mother’s Hospital	883	224 (25.4%)	887	206 (23.2%)	1.09 (0.93 - 1.29)	0.293	
Other Sites	746	247 (33.1%)	750	225 (30.0%)	1.10 (0.95 - 1.28)	0.196	
Cesarean (fetal distress was a factor)							0.859
Mater Mother’s Hospital	883	106 (12.0%)	887	93 (10.5%)	1.14 (0.88 - 1.49)	0.312	
Other Sites	746	127 (17.0%)	750	108 (14.4%)	1.18 (0.93 - 1.50)	0.164	
IVB (fetal distress was a factor)							0.920
Mater Mother’s Hospital	883	118 (13.4%)	887	113 (12.7%)	1.05 (0.82 - 1.33)	0.697	
Other Sites	746	120 (16.1%)	750	117 (15.6%)	1.03 (0.82 - 1.30)	0.797	
Spontaneous vaginal birth							0.450
Mater Mother’s Hospital	883	494 (55.9%)	887	521 (58.7%)	0.95 (0.88 - 1.03)	0.235	
Other Sites	746	370 (49.6%)	750	410 (54.7%)	0.91 (0.82 - 1.00)	0.050	
Post-partum hemorrhage (>1 Liter)							0.036
Mater Mother’s Hospital	880	73 (8.3%)	882	72 (8.2%)	1.02 (0.74 - 1.39)	0.920	
Other Sites	741	91 (12.3%)	745	56 (7.5%)	1.63 (1.19 - 2.24)	0.002	

Legend: IVB, Instrumental Vaginal Birth; *Log-binomial regression model; †Not Done for Cord Artery pH<7 assumed to be No.

Supplementary Table 3: Side-effects potentially related to sildenafil citrate*

	Sildenafil Citrate (N=1552)	Placebo (N=1557)	P-value[†]
Headache			
Yes	3 (0.2%)	7 (0.4%)	0.343
No	1549 (99.8%)	1550 (99.6%)	
Diarrhoea			
Yes	0	1 (0.1%)	0.999
No	1552 (100.0%)	1556 (99.9%)	
Dizziness			
Yes	0	5 (0.3%)	0.062
No	1552 (100.0%)	1552 (99.7%)	
Flushing			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Gastro-oesophageal reflux/dyspepsia			
Yes	2 (0.1%)	1 (0.1%)	0.624
No	1550 (99.9%)	1556 (99.9%)	
Myalgia/limb pain			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Visual changes			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Vomiting			
Yes	0	1 (0.1%)	0.999
No	1552 (100.0%)	1556 (99.9%)	
Bronchitis			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Cellulitis/skin rash			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Epistaxis			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Hypo/paraesthesia			
Yes	1 (0.1%)	1 (0.1%)	0.999
No	1551 (99.9%)	1556 (99.9%)	
Tremor			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	
Vaginal Itch			
Yes	0	0	
No	1552 (100.0%)	1557 (100.0%)	

*Excludes 148 women that received no treatment

†Fisher's exact test.